JUAN CAMILO ZAPATA OROZCO

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PROFESSIONAL SUMMARY

Full Stack and Backend Software Engineer (.NET, Python, and React) with 3+ years of experience designing, developing, and maintaining enterprise web applications, RESTful APIs, microservices, and real-time distributed systems on Linux and Docker. Strong expertise in Hexagonal and Clean Architecture, SOLID principles, Inversion of Control (IoC), Test-Driven Development (TDD), and relational database optimization (SQL Server and PostgreSQL). Hands-on experience implementing and supervising Generative AI and Agentic AI workflows (AI Agents, Azure OpenAI Service, process automation), hardware-software integration (Embedded Linux / Edge), and CI/CD pipelines in GitLab and Microsoft Azure DevOps within agile Scrum teams.

TECHNICAL SKILLS

- **Backend & Programming Languages:** C#, .NET 8, ASP.NET Core Web API, Minimal APIs, Entity Framework Core, ADO.NET, Python 3, FastAPI, Pydantic v2, asyncio, aiohttp, SQLAlchemy, Alembic, SQL, TypeScript, JavaScript (ES6+), Bash
- **Frontend & UI:** React, TypeScript, HTML5, CSS3, reusable component architecture, interactive command-line interfaces (Python Rich CLI)
- **Databases:** SQL Server (T-SQL, Stored Procedures, Views, Indexes, Query Optimization), PostgreSQL, Relational Data Modeling, automated schema migrations (Alembic, EF Core)
- **Architecture & Design Patterns:** Hexagonal Architecture, Clean Architecture, Layered Architecture, Microservices, SOLID Principles, Inversion of Control (IoC / Dependency Injection), GRASP, Domain-Driven Design (DDD)
- **Agentic AI, Generative AI & Automation:** AI Agent supervision and workflow automation in software engineering, Microsoft Azure OpenAI Service (AI-050), Azure AI Fundamentals (AI-900), LLM API integration, Model Context Protocol (MCP), Power Apps
- **Cloud, DevOps & Containers:** Docker, Docker Compose (multi-stage builds), Microsoft Azure (Azure DevOps, Azure Portal IAM/Roles), GitLab CI/CD, Git, GitHub, Linux (Debian/Ubuntu), SSH environment configuration
- **Embedded Systems, Networking & Security:** Embedded Linux Devices (Raspberry Pi / Edge), hardware-software integration (I2C, GPIO), ZeroMQ (distributed messaging and cryptographic authentication), REST APIs, OpenAPI/Swagger, HTTPS/TLS, OAuth2, JWT, Webhooks
- **Testing, Quality & Methodologies:** Test-Driven Development (TDD), Unit, Functional, Integration & Smoke Testing (pytest, unittest, xUnit, Postman), Code Coverage, Static Analysis (pylint, mypy, dotnet format), pre-commit, Husky.NET, Agile Scrum (SFPC™ Certified)

PROFESSIONAL EXPERIENCE

Backend & Full Stack Software Developer (Python / Linux / Architecture & AI) | Intelligent Systems & Real-Time Control Sector | October 2024 - Present (2+ Years)

- Designed, developed, and maintained the core server, backend services, and hardware integration modules for a real-time control system running on Linux and embedded devices using Python, Docker, and distributed messaging.
- Architected and maintained a clean, scalable Hexagonal Architecture applying SOLID principles, Inversion of Control (IoC), and GRASP patterns to decouple domain logic from database and hardware infrastructure.
- Practiced Test-Driven Development (TDD) and automated code quality analysis for every user story, maintaining high unit, functional, and integration test coverage (pytest, Postman) across the application and device communications.
- Streamlined repetitive engineering processes and monitored AI Agent efficiency to automate workflow tasks, generate descriptive user stories, maintain technical documentation, and manage repositories and semantic versioning.
- Developed the server-side data management system implementing Alembic database migrations, security protocols (HTTPS, encrypted distributed messaging, access control), and interactive CLI graphics using the Python Rich library.
- Configured and secured development, testing, and production deployment environments on Linux via SSH and Docker Compose.

Software Developer (.NET / SQL Server / React / Azure Scrum) | International Software Consulting Firm | August 2023 - September 2024 (1+ Year)

- Developed and maintained backend solutions using C#, .NET, and SQL Server for internal and client-facing enterprise applications, ensuring optimal performance through query optimization, technical support, and issue resolution.
- Implemented and delivered frontend user stories in React, enhancing user experience and overall product functionality.
- Designed and executed comprehensive integration test suites using Postman to validate REST APIs and microservices across multiple environments.
- Configured and managed security roles, permissions, and access control (IAM) in Microsoft Azure Portal to safeguard cloud resources.
- Actively participated in Scrum ceremonies (dailies, sprint planning, reviews, and retrospectives —including sessions conducted in English) using Microsoft Azure DevOps.
- Created class diagrams, technical architecture documentation, and user story specifications, while refactoring legacy code to improve long-term maintainability and scalability.

KEY PROJECTS

Real-Time Distributed Control & Monitoring Platform (Edge & Core Server) | Python, Hexagonal Architecture, ZeroMQ, SQLAlchemy, Alembic, Docker, GitLab CI, Embedded Linux

- Containerized microservices platform featuring an OpenAPI-documented REST API, encrypted inter-process messaging, hardware interface control, interactive CLI console, and automated CI/CD pipelines.

Enterprise Layered REST API | C#, .NET 8, ASP.NET Core Web API, Entity Framework, SQL, Docker, GitLab CI, Husky.NET

- Clean layered REST API with Swagger/OpenAPI documentation, multi-stage Dockerfile, GitLab CI pipeline, and automated pre-commit formatting and quality checks.

Electronic Invoicing API Integration Service | C# .NET (Minimal APIs) & Python (FastAPI, Pydantic v2), OAuth2

- Parallel implementations in .NET and Python integrating external electronic invoicing APIs with OAuth2 authentication, token caching, document issuance, and signed PDF download.

Asynchronous Workflow & Webhook Automation System | Python, asyncio, aiohttp, REST APIs, Webhooks

- Asynchronous integration connecting time-tracking APIs, team communication channels, and repository webhooks using a modular layered architecture.

EDUCATION

- **B.S. in Software Engineering (Ingeniería de Software)** | Corporación Universitaria Minuto de Dios (UNIMINUTO) | In Progress
- **Technical Degree in Software Design and Development Assistant** | Centro de Estudios de Sistemas y Desarrollo (CESDE) | 2023 - 2024
- **Technical Degree in Software Design and Development** | Servicio Nacional de Aprendizaje (SENA) | 2020 - 2022

CERTIFICATIONS & PROFESSIONAL TRAINING

- **AI-050T00: Develop Generative AI Solutions with Azure OpenAI Service** | Microsoft | 2024
- **AI-900T00-A: Microsoft Azure AI Fundamentals** | Microsoft | 2023 - 2024
- **Scrum Foundation Professional Certificate (SFPC™)** | CertiProf | 2023

LANGUAGES

- **Spanish:** Native
- **English:** Intermediate (B1/B2) — Experienced in Scrum ceremonies and technical documentation